

LegalEase AI - Deployment Guide

Overview

This document outlines deployment configurations for LegalEase AI, detailing local development deployment as well as production hosting strategies for cloud platforms.

1. Verified Local Deployment

LegalEase AI has been tested and verified locally on Windows using Python 3.14.6:

Verified Local Endpoints:

- **Frontend URL:** `http://localhost:8501`
- **Backend URL:** `http://localhost:8000`
- **Swagger Documentation:** `http://localhost:8000/docs`
- **Health Check:** `http://localhost:8000/health`

2. Production Architecture Considerations

When transitioning LegalEase AI from local execution to production, observe the following architectural recommendations:

Security Hardening

- **Environment Isolation:** Set `GEMINI_API_KEY` via container secrets or secret managers (e.g. AWS Secrets Manager, GCP Secret Manager, Vault).
- **CORS Restricting:** Restrict `allow_origins` in `backend/main.py` from `["*"]` to the exact production frontend domain (e.g. `["https://app.legalease.ai"]`).
- **Rate Limiting:** Attach middleware (e.g. `slowapi`) to prevent API abuse on `/generate`.
- **Request Size Limiting:** Enforce a maximum payload limit (e.g. 500KB) to prevent resource exhaustion.

Performance & Scaling

- **ASGI Concurrency:** Run Uvicorn with multiple worker processes behind a reverse proxy:

```
bash
gunicorn backend.main:app -w 4 -k uvicorn.workers.UvicornWorker --bind 0.0.0.0:8000
```

- **Reverse Proxy:** Deploy Nginx or Cloudflare in front of the application for SSL/TLS termination, HTTP/2 multiplexing, and static caching.
- **Statelessness:** The backend is completely stateless; requests can be load-balanced horizontally across multiple instances or containers.

3. Cloud Deployment Options

The system can be deployed to modern cloud hosting platforms:

Option A: Streamlit Community Cloud (Frontend) + Render / Railway (Backend)

- **Frontend:** Deploy the repository to [Streamlit Community Cloud](#)[🔗]. Set the repository root to `frontend/app.py` and define `BACKEND_URL` in Streamlit secrets.
- **Backend:** Deploy the repository as a Python Web Service on [Render](#)[🔗] or [Railway](#)[🔗]. Set the start command to:

```
bash
uvicorn backend.main:app --host 0.0.0.0 --port $PORT
```

Set `GEMINI_API_KEY` in environment variables.

Option B: Dockerized Container Deployment (Fly.io / Google Cloud Run / AWS ECS)

Create a production `Dockerfile`:

```
dockerfile
FROM python:3.11-slim

WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

COPY . .

EXPOSE 8000
CMD ["uvicorn", "backend.main:app", "--host", "0.0.0.0", "--port", "8000"]
```

Note: While cloud deployment strategies are documented above for future reference, current active verification has been completed for local deployment on `localhost:8000` and `localhost:8501`.